Information Security and Cyber Laws

1. Learning Outcome:
After the completion of the course, the students will be able to develop basic understanding of security, cryptography, system attack and defences against them.

2. Prerequisites: NIL
3. Semester: 6
4. Course Type: Elective
5. Course Level: 300-399
6. Theory Credit: 4
7. Practical Credit: 0
8. No of Hours:

a) Theory: 60 hrs (60 classes)
b) Practical: NIL
c) Non Contact: NIL

9. List of Books:

(a) Merkow, M., & Breithaupt, J.(2005) Information Security Principles and Practices. 5th edition. Prentice Hall.
(b) Cryptography And Network Security Principles And Practice, Fourth or Fifth Edition, William Stallings, Pearson Edition.
(c) Cyber Law & Cyber Crimes, Advocat Prashant Mali; Snow White publications, Mumbai
(d) The Information Technology Act, 2000; Bare Act – Professional Book Publishers, New Delhi

10. Contents of Syllabus:

UNIT 1: Introduction (15 Lectures)

Basic components of security (Confidentiality, Integrity and Availability), Attacks, Computer Crime, Security Services, Security Mechanism, Cyber Crimes, information Technology ACT, Cryptography, Substitution Cipher, Transposition Cipher, Block Cipher, Stream Cipher, Confusion, Diffusion, Symmetric Key, Asymmetric Key, Encryption, DES Algorithm, Hash Function, Digital Signature, Digital Certificate.

UNIT 2: Program Security (10 Lectures)

Program Security, Program Errors, Buffer Oveflow, Incomplete mediation, Time-of-check to Time-of- use Errors, Malicious codes, Virus, Threats, Control against Programs, Program Security Issues. Protection in OS: Memory and Address protection, Access control, File protection, User Authentication.

UNIT 3: Database Security (10 Lectures)

Reliability, Integrity, Sensitive Data, Inference, Multilevel Security,Issues regarding the right to access information: Protecting Data, Multiple security level and categorization of data and users, Loss of integrity, Loss of availability, Loss of confidentiality, Access control, Inference control, flow control, data encryption

UNIT 4: Security in Networks (Cyber Attack) (15 Lectures)

Threats in Networks, Security Controls- Architecture, Encryption, Content Integrity, Strong Authentication, Firewalls: Design and Types of Firewalls, Intrusion Detection System, Secure Email, Denial-of-service attacks, Man in the middle Attack, Phishing, Spoofing and Spam Attacks, Drive-by attack, SQL Injection, Birthday attack, Social Engineering attack, Password Attack. Cross site scripting Attack, Malware Attack, Administering Security, Security Planning, Risk Analysis, Organisational Security Policy, Web Servers and Browsers, HTTP, Cookies, Caching, Secure Socket Layer (SSL), Secure Electronic Transaction (SET), E-mail Risks, Spam, E-mail Protocols, Simple Mail Transfer

Protocol (SMTP), Post office Protocol (POP), Internet Access Message protocol (ICMP), Secured Mail: Pretty Good Privacy (PGP), S/MIME (Secure/Multipurpose Internet Mail Extensions)

UNIT 5: Cyber Laws (10 Lectures)

Cyber crime, Types of crimes, Information technology Act 2000: Salient Feature of IT Act 2000, various authorities under IT Act and their powers, Penalties & Offences, amendments, Sections under the Information Technology Act such as:

- [Section 43] Penalty and compensation for damage to computer etc.
- [Section 65] Penalty for temping with the computers source documents
- [Section 66] Punishment for hacking with computer system, data alteration etc
- [Section 66A] Punishment for sending offensive messages through any communication services
- [Section 66B] Receiving stolen computer's resources or communication devices dishonestly
- [Section 66C] Punishment for identity theft
- [Section 66D] Punishment for cheating by impersonation by using computer resource
- [Section 66E] Punishment for violation of privacy
- [Section 66F] Punishment for cyber terrorism
- [Section 67] Punishment for publishing or transmitting obscene material in electronic form
- [Section 67A] Punishment for publishing or transmitting of material containing sexually explicit act, etc. in electronic form
- [Section 67B] Punishment for publishing or transmitting of material depicting children in sexually explicit act, etc. in electronic form
- [Section 72] Breach of confidentiality and privacy

Particulars of course designer:

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Artificial Intelligence

1. Learning Outcome:
After completing this course, students will know the fundamentals of artificial intelligence (AI), identify problems where artificial intelligence techniques are applicable and able to apply basic principles of AI in solutions that require problem solving, inference, perception, knowledge representation, and learning.

2. Prerequisites: NIL
3. Semester: 6
4. Course Type: Elective
5. Course Level: 300-399
6. Theory Credit: 3
7. Practical Credit: 1
8. No of Hours:

a) Theory: 45 hrs (45 classes)
b) Practical: 30 hrs (15 classes)
c) Non Contact: NIL

9. List of Reference Books:

(a) Rich & Knight, Artificial Intelligence – Tata McGraw Hill, 2nd edition, 1991.

(b) Russell & Norvig, Artificial Intelligence-A Modern Approach, LPE, Pearson Prentice Hall, 2nd edition, 2005.

(c) W.F. Clocksin and Mellish, Programming in PROLOG, Narosa Publishing House, 3rd edition, 2001.

(d) DAN.W. Patterson, Introduction to A.I and Expert Systems – PHI, 2007.

(e) Ivan Bratko, Prolog Programming for Artificial Intelligence, Addison-Wesley, Pearson Education, 3rd edition, 2000.

10. Contents of Syllabus:
A. Theory

UNIT 1: Introduction (4 Hours)
Introduction to Artificial Intelligence, Background and Applications, Turing Test and Rational Agent approaches to AI, Introduction to Intelligent Agents, their structure, behavior and environment.

UNIT 2: Problem Solving and Searching Techniques (16 Hours)
Problem Characteristics, Production Systems, Control Strategies, Breadth First Search, Depth First Search, Hill climbing and its Variations, Heuristics Search Techniques: Best First Search, A* algorithm, Constraint Satisfaction Problem, Means-End Analysis, Introduction to Game Playing, Min-Max and Alpha-Beta pruning algorithms.

UNIT 3: Knowledge Representation (14 Hours)
Introduction to First Order Predicate Logic, Resolution Principle, Unification, Semantic Nets, Conceptual Dependencies, Frames, and Scripts, Production Rules, Conceptual Graphs. Programming in Logic (PROLOG)

UNIT 4: Dealing with Uncertainty and Inconsistencies (6 Hours)
Truth Maintenance System, Default Reasoning, Probabilistic Reasoning, Bayesian Probabilistic Inference, Possible World Representations.

UNIT 5: Understanding Natural Languages (5 Hours)
Parsing Techniques, Context-Free and Transformational Grammars, Recursive and Augmented Transition Nets.

B. Practical:

- Write a prolog program to calculate the sum of two numbers.
- Write a prolog program to find the maximum of two numbers.
- Write a prolog program to calculate the factorial of a given number.
- Write a prolog program to calculate the nth Fibonacci number.
- Write a prolog program, insert_nth(item, n, into_list, result) that asserts that result is the list into_list with item inserted as the nth element into every list at all levels.
- Write a Prolog program to remove the nth item from a list.
- Write a Prolog program, remove nth (Before, After) that asserts the After list is the Before list with the removal of every nth item from every list at all levels.
- Write a Prolog program to implement append for two lists.
- Write a Prolog program to implement palindrome (List).
- Write a Prolog program to implement max(X,Y,Max) so that Max is the greater of two numbers X and Y.
- Write a Prolog program to implement maxlist(List,Max) so that Max is the greatest number in the list of numbers List.
- Write a Prolog program to implement sumlist(List,Sum) so that Sum is the sum of a given list of numbers List.
- Write a Prolog program to implement two predicates evenlength(List) and oddlength (List) so that they are true if their argument is a list of even or odd length respectively.
- Write a Prolog program to implement reverse (List, Reversed List) that reverses lists.
- Write a Prolog program to implement maxlist (List, Max) so that Max is the greatest number in the list of numbers List using cut predicate.
- Write a Prolog program to implement GCD of two numbers.
- Write a prolog program that implements Semantic Networks/Frame Structures.

Particulars of course designer:

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Graph Theory

1. Learning Outcome: After completing this course, students will have understanding of graph theoretic concepts, problems and associated algorithmic solutions.
2. Prerequisites: NIL
3. Semester: 6
4. Course Type: Elective
5. Course Level: 300-399
6. Theory Credit: 4
7. Practical Credit: 0
8. No of Hours:

a) Theory: 60 hrs (60 classes)
b) Practical: NIL
c) Non Contact: NIL

9. List of Books:

(a) Introduction to Graph Theory, Douglas B. West, Pearson

(b) Introduction to Graph Theory, Robin J. Wilson, Pearson Education Limited

(c) Graph Theory with Applications to Engineering and Computer Science, Narasingh Deo, PHI

10. Contents of Syllabus:

Unit I: Introduction **5 hrs**

Graph, directed and undirected graph, weighted and unweighted graph, simple and multigraph, degree, in degree and out degree, Handshaking theorem, complete graph, bipartite graph, cut set, cut vertices, graph representations: incidence matrix, adjacency matrix and adjacency list, BFS traversal and DFS traversals on a graph using stack and queue data structures, isomorphism, homomorphism

Unit II: Connectivity, paths and cycle **15 hrs**

Walk, path and cycle, connected graphs, disconnected graphs, components, Hamiltonian path, Hamiltonian cycle, Hamiltonian graphs, Dirac's theorem, Eulerian path, Eulerian cycle, Euler graphs, Fleuri's algorithm, 2-connected graphs, connectivity and digraph, k-connected and k-edge connected graphs, application of Menger's theorem, Shortest path problem, variations of shortest path problem: single source shortest path problem, single pair shortest path problem and all pairs shortest path problem, Dijkstra's algorithm, Bellman Ford algorithm, Floyd Warshall's algorithm, Johnson's algorithm

Unit III: Tree **12 hrs**

Tree, forest, properties of tree, spanning tree, spanning forest, counting trees, Cayley's theorem, matrix-tree theorem, minimum spanning tree, Kruskal's algorithm, Prim's algorithm, disjoint spanning trees, graph decomposition, graceful labeling, graceful graph, binary tree, binary search tree, AVL tree, multiway search tree, B tree, B+ tree

Unit IV: Matching and coloring **13 hrs**

Matching, bipartite matching, maximum bipartite matching, Ford Fulkerson's algorithm for finding maximal bipartite matching, perfect bipartite matching, non-bipartite matching, maximal non-bipartite matching, largest maximal matching, perfect non-bipartite matching, Hall's Marriage theorem, vertex cover, vertex cover and matching, independent sets, dominating sets, atable matching, Hungarian algorithm, introduction to Edmonds Blossom shrinking algorithm, vertex

coloring, k-colorable graph, chromatic number, Brook's theorem, clique number, map coloring problem

Unit V: Digraph **7 hrs**

Digraph, simple digraph, connected and strongly connected digraph, orientable graph, Eulerian digraph, Hamiltonian digraph, tournament, Markov chains, Flow networks, residual graph, augmenting path, Ford Fulkerson's algorithm

Unit VI: Classical problems **8 hrs**

Travelling Salesman Problem, variants of Travelling Salesman Problem, Chinese Postman Problem, variants of Chinese Postman Problem, the minimum connector problem, Huffman coding and Huffman tree, Konisgsberg bridge problem, three utilities problem

Particulars of course designer:

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